



Speed-FairMOT: multi-class multi-object tracking for real-time manipulation

Cheng Ju¹ · Ziran Li^{1,2} · Koki Terakado¹ · Akio Namiki¹

Received: 16 January 2025 / Revised: 28 May 2025 / Accepted: 13 June 2025 / Published online: 2 July 2025
© The Author(s) 2025

Abstract

In order to achieve complex tasks at high speed in robot manipulation, the ability to perform multi-object tracking (MOT), which recognizes the many objects in the surrounding area using camera-based real-time image data processing, is essential. To overcome traditional MOT methods slow tracking speeds challenges, we propose Speed-FairMOT, a deep-learning based real-time multi-class MOT method. We evaluate the Speed-FairMOT on MOT17 dataset and our custom synthetic dataset, achieving over 41% improvement in speed and slightly decrease of 5.9% in tracking performance as trade-off compared to the original FairMOT. We verified proposed Speed-FairMOT using a camera mounted on a robot manipulator as a hand-eye system. As the result, we were able to achieve MOT at an maximum speed over 58 fps in real-time. This real-time speed is sufficient for feedback control in robotic manipulation system.

Keywords Multi-Object Tracking · Real-Time Application · Deep Learning · Hand-eye Manipulation

1 INTRODUCTION

With the growing popularity of robotic manipulation systems in recent years, the operation environment has become more general and complex. Typical working scenarios, such as medical applications, agriculture, and home usage, demand a high level of environment perceptual ability. Image data is particularly effective for such environmental recognition around the robot due to its flexibility and accessibility [1–4]. Moreover, the capacity to process captured image data and extract environmental information in real-time during

those robotic manipulation process is required. In particular, multi-object tracking (MOT) is a very effective method for environment information processing in such manipulation tasks shown as Figure 1. Robot motion is performed based on information about the objects in surrounding environment that have been processed by MOT method. However, due to the heavy processing load and slow tracking speed, conventional MOT techniques are often limiting their effectiveness in real-time robot manipulation controls such as visual servoing, obstacle avoidance and object grasping. As a result, tracking becomes a bottleneck in the system.

To address this challenge, we developed a high-speed MOT method called Speed-FairMOT specifically for real-time robotic control applications. Our proposed Speed-FairMOT is built on top of FairMOT, an anchor-free deep-learning MOT method. Speed-FairMOT balances between tracking accuracy and inference speed, enabling real-time multi-class MOT. We utilize A lightweight architecture called ShuffleNetV2 [5] as the backbone, and separate feature fusion parts called DLA Fusion in modified DLA34 from CenterNet [6] as the neck network.

To evaluate the proposed model, we select the public dataset MOT17 [7] as our primary benchmark with a result showing acceptable tracking accuracy at real-time. Furthermore, for real-world applications, we have implemented training and evaluation using a custom synthetic dataset

Z. Li, K. Terakado, A. Namiki: These authors contributed equally to this work.

✉ Cheng Ju
JuchengJC@outlook.com

Ziran Li
nature_lucky@163.com

Koki Terakado
23wm4130@student.gs.chiba-u.jp

Akio Namiki
namiki@faculty.chiba-u.jp

¹ Graduate School of Engineering, Chiba University, Chiba City, Chiba Ken, Japan

² School of Automation, Wuxi University, Wuxi City, Jiangsu Province, China

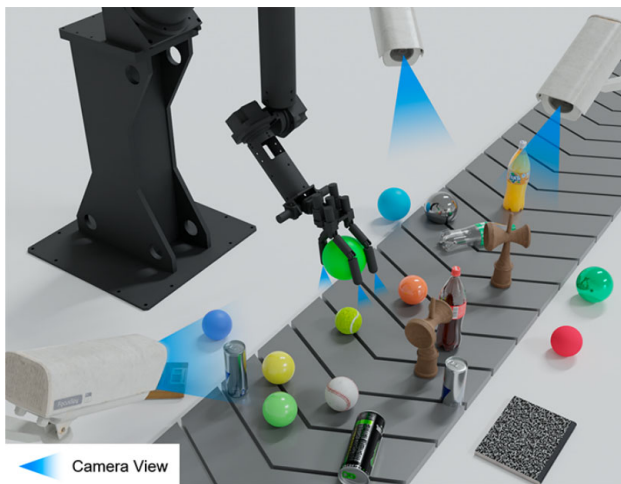


Fig. 1 Example scenario of robotic manipulation with MOT. MOT methods can enhance precision in tasks such as object sorting, assembly, and hand-eye manipulation. Slow MOT method become the bottleneck of the whole system

which is specific designed for real-world multi-class MOT tasks. Finally, We deployed Speed-FairMOT within our robot manipulation system to evaluate its performance for real-time multi-class MOT tasks in an actual system.

Our contributions are as follows:

1. We proposed Speed-FairMOT, a multi-class MOT method specific for real-time vision-based applications.
2. We verified the efficiency of the proposed Speed-FairMOT on public dataset MOT17 and custom synthetic dataset compared with the original FairMOT.
3. We implemented Speed-FairMOT on our robotic hand-eye system. Speed-FairMOT accomplished MOT task and achieved an inference speed of 41 FPS average and 58 FPS maximum.

2 Related work

The existing MOT methods can be classified into two categories: separate models and joint models.

2.1 Separate detection and tracking

Current best performed multi-object tracking methods are mainly as separate detection and tracking paradigm. Those trackers primarily follow a two-step workflow [8]. First step is using a object detector to recognize all objects of interest in frames. Second step is using a separate tracker to associate these previously detected objects by motion information or appearance information. This separation allows for more tailored optimization of each model, potentially improving the overall tracking performance.

For detection part, popular methods such as Faster R-CNN [9], SDP [10] and YOLO-based network [11] are widely used. Tracking part is the primary focus for most existing works. DeepSORT [12] based on SORT [13], applies deep convolutional network to extract appearance features for association. ByteTrack [14] enhances tracking performance by reusing low-confidence detections based on YOLOX [11] detector.

However, implement speed of such models is usually considered very slow due to that two stages need to be accomplished individually in series.

2.2 Joint detection and embedding

Joint Detection and Embedding (JDE) [15] is a category of deep learning based multi-objects tracking methods which perform object detection and embedding feature extraction in a single network in order to reach a faster inference speed. Embedding features can be extracted as re-ID appearance features or objects motion features and then be used in association. The original JDE is based on the YOLOv3 [16] backbone using re-ID as embedding achieves a near real-time runtime. CenterTrack [17] with pair inputs learn a 2D offset between two adjacent frames and associate them by center distance. FairMOT [18] uses a anchor-free center-based object detection framework to mitigate the unfairness problem caused by anchors. All JDE models can achieve a relative fast tracking speed in deep learning based methods. However, these models typically has a lower accuracy compared to separate detection and tracking methods considering as a trade off of high inference speed. It's crucial to balance between performance and efficiency.

This study work belongs to JDE category architecture. Specifically, proposed Speed-FairMOT takes the advantages of JDE type methods and focus on the inference speed with an acceptable tracking performance.

3 Materials and methods

Figure 2 shows the overall process of the proposed Speed-FairMOT in this paper. Following the original FairMOT design, we proposed a combined backbone and neck network, which consists of a features-extracting and a refining network.

3.1 Backbone and neck

The original FairMOT uses an enhanced DLA network as a single encoder-decoder network, which consists of ResNet-34 network to extract features as backbone stages. The enhanced DLA network has more skip connections between low-level and high-level features which is similar to the Fea-

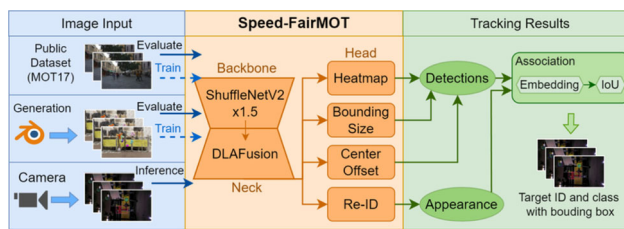


Fig. 2 Structure of Speed-FairMOT process. We propose a real-time specific multi-class multi-object tracking method. And verify this method on public dataset MOT17. For real-world application, we present a dataset generation tool and train Speed-FairMOT on custom generated dataset. Finally we test proposed method Speed-FairMOT on real-world footage

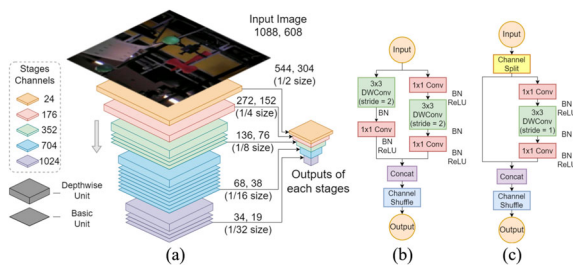


Fig. 3 (a) Structure of ShuffleNetV2 adopted in Speed-FairMOT as backbone. (b) Depthwise unit structure used in backbone. (c) Basic unit structure used in backbone

ture Pyramid Network(FPN) [19]. Additionally, deformable convolution is used as replacement of regular convolution layers in all up-sampling modules since they can dynamically adjust the receptive field according to object scales and poses. However, enhanced DLA is computationally heavy and not suitable for real-time computation.

In this study, ShuffleNetV2 is adopted as backbone to achieve a higher speed with acceptable accuracy. Like the original Resnet [20], ShuffleNetV2 has the same residual structure as basic unit. Structure is shown in Figure 3. It introduces channel split operation at the beginning of each unit and channel shuffle operation at the end of each unit, which are beneficial to the performance without impacting too much to running speed. We separate feature fusion parts from enhanced DLA34 network as neck, called DLA-Fusion. Figure 4 shows its structure. This neck network utilizes the Iterative Deep Aggregation(IDA) from the original DLA for interpolation and a further augmentation to reach the necessary output resolution for the task. Additional skip connections from bottom layers and deformable convolution layer in each upsampling stages help to achieve a high performance. The output shape of the neck network keeps the same as the original FairMOT decoder network. With input image size width W and height H of backbone network, the feature map of neck network has the shape $C \times \frac{W}{4} \times \frac{H}{4}$.

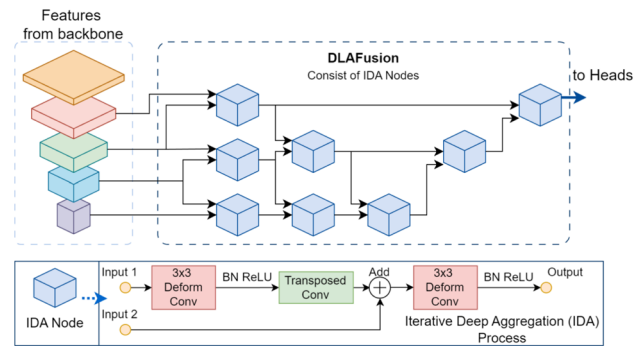


Fig. 4 Structure of DLA-Fusion in Speed-FairMOT as neck. We separate feature fusion parts from enhanced DLA34 network called DLA-Fusion. The basic unit of IDA node in DLA-Fusion is shown

4 Custom synthetic datasets generation

To provide a training source for our proposed Speed-FairMOT, we generated a custom synthetic dataset specific for MOT tasks in our hand-eye system environment as training source.

Synthetic dataset is a collection of data that is artificially generated to mimic real-world data rather than being directly produced by humans [21]. Such datasets can be tailored to meet the specific requirements which may not be included in real-world datasets, such as certain object categories, surface materials and environment. High quality unbiased labels can be provided for many image based tasks such as semantic, text localization, object detection and classification, thus it has been used effectively for training [22]. On the other hand, Synthetic datasets often suffer from a lack of realistic variations, such as complex lighting conditions, occlusions, and dynamic backgrounds. Furthermore, the synthetic-to-real adaptation remains a significant challenge, as models trained exclusively on synthetic data frequently encounter substantial performance drops when applied to real domains due to the domain gap in visual appearance and statistical distribution. Even with imperfect realism, synthetic data is reported as an effective tool.

In this study, we created an add-on for Digital Content Creation(DCC) Blender [23], an open source 3D digital content creation tool which supports the entirety of the 3D pipeline, to achieve procedural datasets generation. An overview of our generation pipeline is shown as Figure 5. All image sequences used in this study are rendered by Cycles, a ray-trace based renderer engine inside Blender, to provide realistic lighting condition. The final dataset used in this study contains total 75 sequences with 150 frames per sequence for training and additional 5 sequences with 100 frames per sequence for validation. All sequence are rendered by 120 FPS to mimic a relative high speed camera.

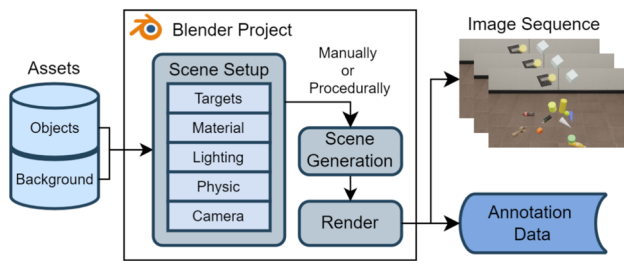


Fig. 5 Our dataset generation pipeline overview. We generated photo-realistic image sequences by any given meshes and HDRI background. High quality annotations are also provided

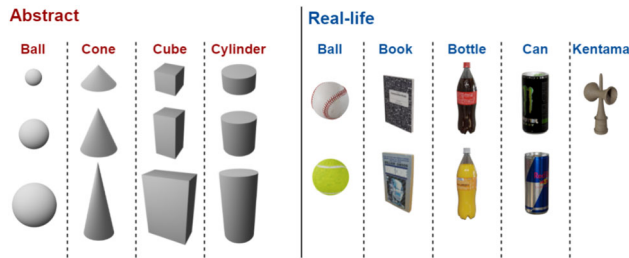


Fig. 6 Object mesh assets and render example in this study. All object mesh assets used in this study are divided into real-life objects and abstract object

4.1 Dataset object types

Considering the convenience of real-world experiments and the variety of objects, 5 real-world daily objects categories with certain material challenges and 5 basic shape abstract objects categories are used, which are shown in Figure 6. All 5 real-world objects models are handmade and matched to provide the complexity in real-world objects. Also, blender built-in physic simulation is enable on all objects to achieve a realistic scene. Additionally, 5 moving camera paths and 3 fixed camera positions are provided.

4.2 Dataset environment

Both High Dynamic Range Image (HDRI) and hand-built environment are used.

HDRIs capture a wide range of luminance between the brightest and darkest areas, enabling photo-realistic lighting quality. In this study, we used 10 CC0 licensed open-source HDRI files as a pool to provide complex lighting conditions during datasets generation process. Besides using HDRI images only as the background of rendering, we included a manually modeled environment with scanned data based on our real-world experiment environment and manually matched lighting situation. This handmade scene was included in training datasets to reduce gap between training synthetic domain and real-world experiment domain.

Scanned data was obtained by a commercial 3D asset scanning mobile app called RealityScan by Epic Games Inc. [24]

5 Experiments

The entire experiment was divided into three steps. The first step involved evaluating our Speed-FairMOT on the public dataset MOT17 to serve as the primary performance assessment. On the second we trained and assessed our model on a custom synthetic dataset designed to simulate real-world conditions before deployment in an actual system. To validate the simulation results obtained in the second step, the third step consisted of demonstrating the proposed Speed-FairMOT on our real-world hand-eye system.

5.1 Datasets and metrics

We evaluated the proposed Speed-FairMOT on 2 datasets: MOT17 [7] and custom generated datasets. MOT17 training dataset is divided into training half and valid half. MOT17 dataset is used for testing single class multi-objects tracking performance and Custom generated dataset is used for multi-classes multi-objects tracking. The CLEAR metrics [25] are used for evaluating overall tracking accuracy.

5.2 Implementation details

Both training and evaluating were implemented on single NVIDIA Geforce RTX3090 GPU and Intel Core i9-12900k CPU. All models are trained for 60 epochs on Adam optimizer with a starting learning rate of $1e^{-4}$ and the batch size is set to be 12. The learning rate decays to $1e^{-5}$ at the 20th epoch. Scale augmentation is used as data augmentation for all training.

5.3 Comparison results on MOT17

5.3.1 Backbone network structure

First we compared ShufflenetV2 backbone network with original DLA34 backbone. ShufflenetV2 provides several levels of complexities to adapt different working environment. In this experiment, we tested $1.0\times$, $1.5\times$ and $2.0\times$ those 3 levels of complexities. All the difference between those 3 levels is listed in Table 1. For further general validation, we also provided 3 common lightweight backbone comparison results which were MobileNetv3-Small [26], ResNet18 [20] and VGG11 [27]. All networks used the same DLA Fusion neck.

Table 2 shows the comparison results. Among all 7 models, the original DLA34 has the best performance but the slowest speed, 60.923 MOTA and 44.28 FPS. All 3 lev-

Table 1 Overall architecture of ShuffleNetV2 in 3 different complexities levels

Layer	Output channels		
	1×	1.5×	2×
Image	3	3	3
Conv1 MaxPool	24	24	24
Stage2	116	176	244
Stage3	232	352	488
Stage4	464	704	976
Conv5	1024	1024	2048

Table 2 Backbone comparing results

Backbone	MOTA↑	IDF1↑	IDs↓	FPS↑
DLA34	60.923	66.104	528	44.28
ShuffleNetV2-1.0x	53.642	60.388	590	70.75
ShuffleNetV2-1.5x	57.300	63.298	537	62.86
ShuffleNetV2-2.0x	57.292	63.032	517	55.94
MobileNetV3-Small	46.288	56.602	512	88.25
ResNet18	49.682	58.666	551	72.88
VGG11	56.754	64.145	626	29.94

The best results except the original FairMOT are shown in bold

els ShuffleNetV2 models shows certain trad-off. Level 1.0× model has the best inference speed 70.75 FPS with 59% increase, but the tracking performance decays too much which is 53.642 with 7.281 decrease. Level 1.5× and 2.0× model has the same tracking performance but 2.0× model is noticeable slower. Level 1.5× keeps a good balance between tracking performance and inference speed, 57.3 MOTA with 3.623 decrease and 62.86 FPS with 41% increase. We conclude that ShuffleNetV2 is regarded as a valuable performance trade-off for speed. ShuffleNetV2 backbone network performance will be saturated at 1.5× level.

5.3.2 Neck network structure

Since this DLAfusion neck network from enhanced DLA-34 shares similar structures with Path Aggregation Network(PAN) [28], we compared it with a PAN structure network to verify the efficiency of DLAfusion. Ghost-PAN [29] as a lightweight high efficient PAN feature fusion network with Ghost module [30] is selected. We also provided traditional neck networks as comparison references, which were Linear FPN [19], Linear PAN [28] and PAN without Ghost modules in GhostPAN. All neck networks were tested on DLA34 and ShuffleNetV2 backbone to remove the contingency and extend the validity of efficiency. ShuffleNetV2 includes 1.5× and 2.0× level in this section.

The results are shown in Table 3. With GhostPAN as the neck network, all models have a significant track-

Table 3 Neck comparing result

Backbone	MOTA↑	IDF1↑	IDs↓	FPS↑
Neck: DLAfusion				
DLA34	60.923	66.104	528	44.28
ShuffleNetV2-1.5x	57.300	63.298	537	62.86
ShuffleNetV2-2.0x	57.292	63.032	517	55.94
Neck: GhostPAN				
DLA34	56.896	65.568	583	69.91
ShuffleNetV2-1.5x	49.869	59.769	700	104.91
ShuffleNetV2-2.0x	51.356	58.972	659	97.2
Neck: Linear FPN				
DLA34	32.477	46.014	537	82.12
Neck: Linear PAN				
DLA34	34.779	46.576	506	81.86
Neck: GhostPAN without Ghost modules				
DLA34	57.283	65.79	598	60.01

The best results except the original FairMOT are shown in bold

ing speed boost. On the other hand, The deterioration of tracking performance became unacceptable specially with ShuffleNetV2 backbone. DLAfusion neck is identified as a efficient network for features fusion in all models. With a close tracking performance and speed result, ShuffleNetV2 at 1.5× level with DLAfusion model shows a leading edge in ID switches(IDs) compared to DLA34 with GhostPAN model which is 537 to 583.

5.4 Validation on custom synthetic dataset

We evaluated proposed Speed-FairMOT and the original FairMOT on our custom generated dataset. We use digitally recreated real-world experiment environment and indoor HDRI to mimic various real-world complex environment. This experiment is also can be recognized as a evaluation on multi-class multi-object online tracking, since MOT17 dataset only focuses on human category. Example results images are shown in Figure 7

We validated all models in the comparison study above on MOT17 dataset to verify the results. Table 4 shows the results in 'Ball' and 'Can' 2 example object categories. Model ShuffleNetV2 at 1.5× level with DLAfusion as neck shows similar even better tracking performance in a noticeable higher inference speed compared to the original FairMOT. Comparing with the original FairMOT, ShuffleNetV2 at 1.5× level with DLAfusion has 67.976 MOTA on "Ball" category with 7.269 increase and 91.942 MOTA on "Can" category with 1.189 decrease. And model ShuffleNetV2 at 1.5× level with DLAfusion has over 44 FPS inference time, 27.86% faster than the original FairMOT which is about 35 FPS.

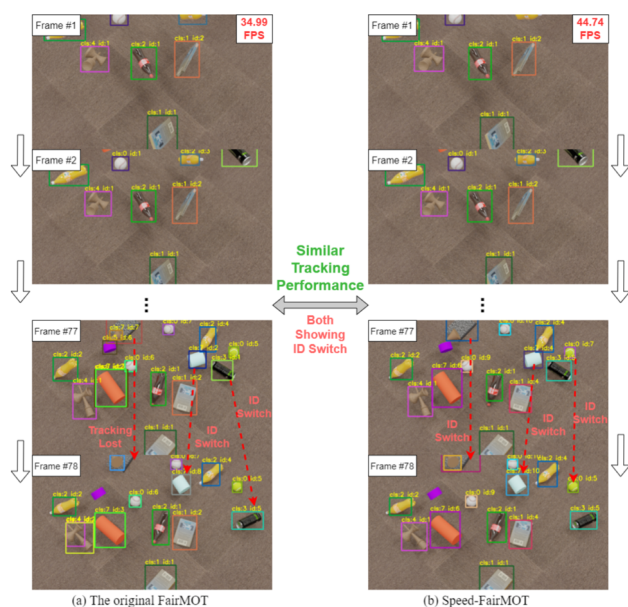


Fig. 7 Custom synthetic dataset experiment result examples. (a) The original FairMOT with tracking speed as average 34.99 FPS. (b) Speed-FairMOT with a higher speed as average 44.74 FPS. Both models exhibited similar tracking performance. However, both also demonstrated instances of ID switches, which could potentially impact subsequent tracking-based tasks

Table 4 Tracking results in example categories on custom dataset

Backbone	MOTA↑	IDF1↑	IDs↓	FPS↑
Neck: DLA Fusion				
Ball				
DLA34	60.707	58.988	44	34.99
ShuffleNetV2-1.5x	67.976	72.088	35	44.74
Can				
DLA34	93.131	84.528	22	34.99
ShuffleNetV2-1.5x	91.942	82.100	23	44.74
Neck: GhostPAN				
Ball				
DLA34	60.314	53.702	45	48.95
ShuffleNetV2-1.5x	50.491	51.699	47	61.44
Can				
DLA34	87.583	79.450	26	48.95
ShuffleNetV2-1.5x	87.054	75.642	22	61.44

Based on results above, we choose ShuffleNetV2 at 1.5× level with DLA Fusion as backbone and neck in Speed-FairMOT network.

Additionally, we presented tracking validation results across four different synthetic dataset domains in Table 5. Both models demonstrated sensitivity to the complexity of scene object distributions, while exhibiting minimal impact from variations in lighting conditions. The proposed Speed-FairMOT exhibits slightly decrease of 5.9% on tracking

Table 5 Tracking results in different domains on custom dataset. All domains target class were set as Kendama

Model	MOTA↑	IDF1↑	IDs↓	FPS↑
Cluttered Background				
FairMOT	64.052	65.910	23	45.68
Speed-FairMOT(Ours)	64.705	59.225	22	67.79
Sparse Background				
FairMOT	78.813	83.312	7	46.15
Speed-FairMOT(Ours)	75.000	83.986	7	66.14
Low Lighting				
FairMOT	92.132	70.309	21	44.56
Speed-FairMOT(Ours)	89.763	72.837	15	66.50
Normal Lighting				
FairMOT	86.816	90.091	3	43.95
Speed-FairMOT(Ours)	88.359	90.105	2	66.58

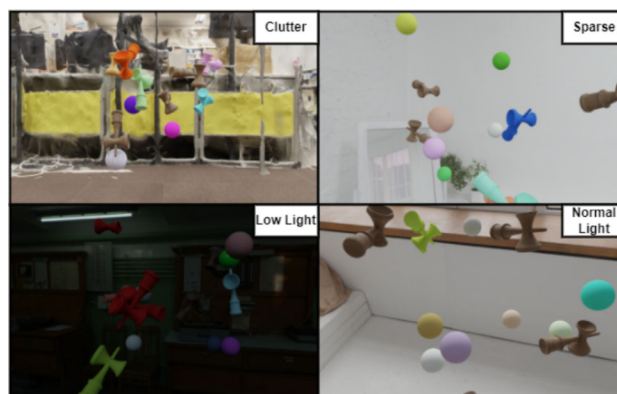


Fig. 8 Custom synthetic dataset examples in different domain

performance in terms of overall accuracy and ID switches as trade-offs, while achieving a over 41% improvement on speed. Both models had a limitation on ID switches. The presence of several ID switches in both models as shown in Figure 7 could adversely affect downstream tasks, such as robotic arm manipulation.

5.5 Real-world experiment

Finally, we deployed Speed-FairMOT network on our robotic multi-vision hand-arm to implement an online experiment. Notably, vision control was not involved in our experiment control strategy, meaning that the tracking methods did not influence the control outcomes. This real-world experiment were conducted solely to validate the real-time tracking capability of our proposed Speed-FairMOT under practical application scenarios.

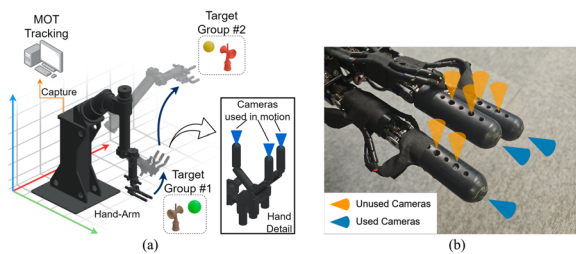


Fig. 9 Experiment platform. Multi-vision hand-eye system was built in previous study. (a) This experiment environment setup and robotic hand movement sequence. (b) Hand setup details used in this experiment

5.5.1 Experiment platform

The experiment platform configuration and setup are shown as Figure 9. The multi-vision hand-arm system consists of a multi-fingered hand which a number of small cameras attached on. The cameras used are IU23N2-Z developed by Sony Inc. with dimensions 2.6 mm × 3.3 mm × 2.32 mm. In this experiment, only vision system are involved. The motion of hand arm is in accordance with the predetermined trajectory. We captured footage from cameras on the finger tips respectively with 640 width pixels and 400 height pixels frame size at 120 FPS. Then we chose 1 camera footage which contain all target objects for this experiment. All models were tested on the same captured footage.

5.5.2 Experiment results

Example results images are shown in Figure 10. According to the comparison, Speed-FairMOT has a similar tracking result as the original FairMOT. Both of two models can accomplish tracking task.

About inference time, results are shown in Table 6. Speed-FairMOT can achieve a tracking speed as over 58 FPS maximum, average 41 FPS. And the original FairMOT can only implement at a speed as 38 FPS maximum, average 29 FPS. Speed-FairMOT is 44.88% faster than the original FairMOT. As the results, proposed Speed-FairMOT network can achieve real-time multi-class multi-object tracking task in real-world system. Higher tracking speed can help improving system controlling precision and response latency by a more frequent target position feedback. However, both models retain limitations from the synthetic dataset domain. In particular, ID inconsistencies may compromise the reliability of downstream tasks, such as high-speed robotic manipulation, representing a limitation for both the original FairMOT and Speed-FairMOT.

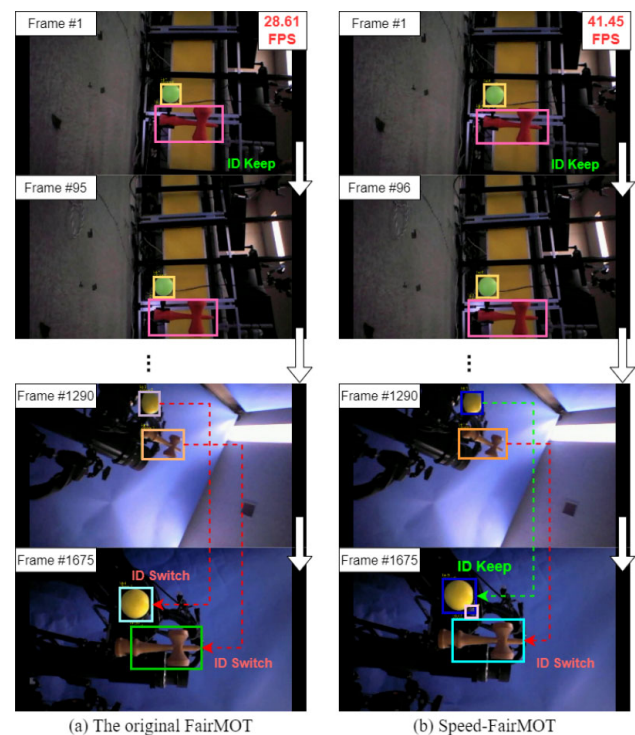


Fig. 10 Hand-eye system experiment result examples. (a) The original FairMOT average 28.61 FPS. (b) Speed-FairMOT average 41.45 FPS. Speed-FairMOT demonstrates similar tracking performance with a substantially higher tracking speed. Nevertheless, both models remain the limitations from the synthetic dataset domain. Furthermore, we have successfully valid our method for deployment in real-world scenarios

Table 6 Backbone comparing results in real-world condition

Model	Max FPS↑	Average FPS↑
FairMOT	38.89 (baseline)	28.61 (baseline)
Speed-FairMOT	58.35 (+50.03%)	41.45 (+44.88%)

The best results are shown in bold

6 Conclusion

In this paper, we proposed Speed-FairMOT, a real-time multi-class multi-object tracking method to improve the perception of vision-based control systems. The key innovation of Speed-FairMOT is its architecture design, which is specialized for high-speed processing. As trade-offs between speed and accuracy, we validated the structures of Speed-FairMOT for significantly higher speed with minimal compromise in tracking performance.

We first evaluated the performance of Speed-FairMOT on the MOT17 public dataset. As a result, Speed-FairMOT achieves 57.3 MOTA and 62.86 FPS on the MOT17 dataset, representing a slightly decrease of 5.9% in MOTA but a significantly increase of 41% in speed compared to the original FairMOT. We next evaluated our custom synthetic dataset which is designed to simulate real-world conditions. Our

Speed-FairMOT performed tracking on 8 categories simultaneously at speed 44.74 FPS, 27.86% faster than the original FairMOT. Moreover, we demonstrated the deployment of Speed-FairMOT on our actual robotic manipulation system. Our Speed-FairMOT proves to successfully accomplish MOT task with a speed of 41 FPS average and 58 FPS maximum, 44.88% faster than the original FairMOT.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11760-025-04395-y>.

Author Contributions C.J. and Z.L. proposed methodology. C.J. and K.T. performed experimental design, data collection, and analysis of results. C.J. and A.N. wrote and checked the manuscript. All authors contributed to the study conception and design.

Funding This work was supported by the Japan Science and Technology Agency Support for Pioneering Research Initiated by the Next Generation (JST SPRING) under Grant JPMJSP2109 and Japan Society for the Promotion of Science (JSPS KAKENHI) under Grant Number JP22H01439.

Data Availability The primary code of this work is publicly available at <https://github.com/Namiki-Laboratory-Chiba-Univ/Argos> and generation tool is publicly available at <https://github.com/Namiki-Laboratory/Prometheus>.

Declarations

Conflicts of Interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Qi, J., Ma, L., Cui, Z., Yu, Y.: Computer vision-based hand gesture recognition for human-robot interaction: a review. *Complex & Intelligent Systems* **10**(1), 1581–1606 (2024)
2. Oka, T., Komura, N., Namiki, A.: Ball juggling robot system controlled by high-speed vision. In: 2017 IEEE International Conference on Cyborg and Bionic Systems (CBS), pp. 91–96 (2017). IEEE
3. Sato, M., Takahashi, A., Namiki, A.: High-speed catching by multi-vision robot hand. In: 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 9131–9136 (2020). IEEE
4. Zhao, Z., Mineshita, H., Lim, H.-o.: Development of a voice information presentation guidance system (vipgs) using coanda-drone-v2. *IEEE Transactions on Electrical and Electronic Engineering*
5. Ma, N., Zhang, X., Zheng, H.-T., Sun, J.: Shufflenet v2: Practical guidelines for efficient cnn architecture design. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 116–131 (2018)
6. Duan, K., Bai, S., Xie, L., Qi, H., Huang, Q., Tian, Q.: Centernet: Keypoint triplets for object detection. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 6569–6578 (2019)
7. Dendorfer, P., Rezaatofghi, H., Milan, A., Shi, J., Cremers, D., Reid, I., Roth, S., Schindler, K., Leal-Taixé, L.: Mot20: A benchmark for multi object tracking in crowded scenes. *arXiv preprint arXiv:2003.09003* (2020)
8. Voigtlaender, P., Krause, M., Osep, A., Luiten, J., Sekar, B.B.G., Geiger, A., Leibe, B.: Mots: Multi-object tracking and segmentation. In: Proceedings of the IEEE/cvf Conference on Computer Vision and Pattern Recognition, pp. 7942–7951 (2019)
9. Ren, S., He, K., Girshick, R., Sun, J.: Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence* **39**(6), 1137–1149 (2016)
10. Yang, F., Choi, W., Lin, Y.: Exploit all the layers: Fast and accurate cnn object detector with scale dependent pooling and cascaded rejection classifiers. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2129–2137 (2016)
11. Ge, Z., Liu, S., Wang, F., Li, Z., Sun, J.: YoloX: Exceeding yolo series in 2021. *arXiv preprint arXiv:2107.08430* (2021)
12. Wojke, N., Bewley, A., Paulus, D.: Simple online and realtime tracking with a deep association metric. In: 2017 IEEE International Conference on Image Processing (ICIP), pp. 3645–3649 (2017). IEEE
13. Bewley, A., Ge, Z., Ott, L., Ramos, F., Upcroft, B.: Simple online and realtime tracking. In: 2016 IEEE International Conference on Image Processing (ICIP), pp. 3464–3468 (2016). IEEE
14. Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., Wang, X.: Bytetrack: Multi-object tracking by associating every detection box. In: European Conference on Computer Vision, pp. 1–21 (2022). Springer
15. Wang, Z., Zheng, L., Liu, Y., Li, Y., Wang, S.: Towards real-time multi-object tracking. In: European Conference on Computer Vision, pp. 107–122 (2020). Springer
16. Redmon, J., Farhadi, A.: YoloV3: An incremental improvement. *arXiv preprint arXiv:1804.02767* (2018)
17. Zhou, X., Koltun, V., Krähenbühl, P.: Tracking objects as points. In: European Conference on Computer Vision, pp. 474–490 (2020). Springer
18. Zhang, Y., Wang, C., Wang, X., Zeng, W., Liu, W.: Fairmot: On the fairness of detection and re-identification in multiple object tracking. *International Journal of Computer Vision* **129**, 3069–3087 (2021)
19. Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2117–2125 (2017)
20. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778 (2016)
21. Liu, R., Wei, J., Liu, F., Si, C., Zhang, Y., Rao, J., Zheng, S., Peng, D., Yang, D., Zhou, D., et al.: Best practices and lessons learned on synthetic data for language models. *arXiv preprint arXiv:2404.07503* (2024)
22. Dosovitskiy, A., Fischer, P., Ilg, E., Hausser, P., Hazirbas, C., Golkov, V., Van Der Smagt, P., Cremers, D., Brox, T.: FlowNet: Learning optical flow with convolutional networks. In: Proceed-

- ings of the IEEE International Conference on Computer Vision, pp. 2758–2766 (2015)
23. Community, B.O.: Blender - a 3D Modelling and Rendering Package. Blender Foundation, Stichting Blender Foundation, Amsterdam (2018). Blender Foundation. <http://www.blender.org>
 24. Epic Games, I.: RealityScan (2023). <https://www.unrealengine.com/en-US/realityscan>
 25. Bernardin, K., Stiefel, R.: Evaluating multiple object tracking performance: the clear mot metrics. EURASIP Journal on Image and Video Processing **2008**, 1–10 (2008)
 26. Howard, A., Sandler, M., Chu, G., Chen, L.-C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., Le, Q.V., Adam, H.: Searching for MobileNetV3 (2019). <https://arxiv.org/abs/1905.02244>
 27. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. arXiv preprint [arXiv:1409.1556](https://arxiv.org/abs/1409.1556) (2014)
 28. Liu, S., Qi, L., Qin, H., Shi, J., Jia, J.: Path aggregation network for instance segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 8759–8768 (2018)
 29. RangilYu: NanoDet-Plus: Super fast and high accuracy lightweight anchor-free object detection model. <https://github.com/RangilYu/nanodet> (2021)
 30. Han, K., Wang, Y., Tian, Q., Guo, J., Xu, C., Xu, C.: Ghostnet: More features from cheap operations. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1580–1589 (2020)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com